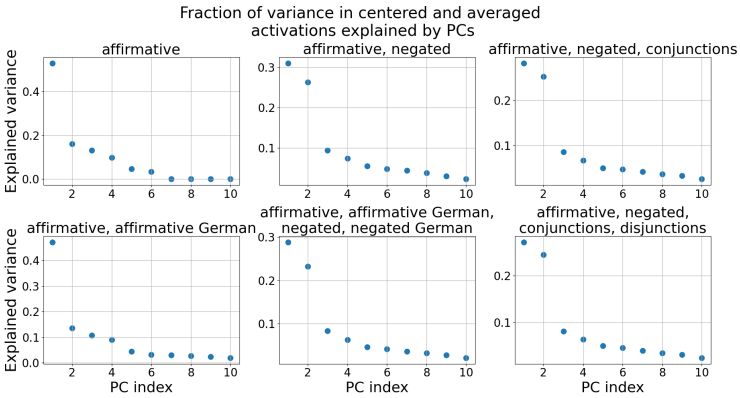


Figure 36: LLaMA3-8B-base: The fraction of variance in the centered and averaged activations $\tilde{\boldsymbol{\mu}}_i^+$, $\tilde{\boldsymbol{\mu}}_i^-$ explained by the Principal Components (PCs). Only the first 10 PCs are shown.

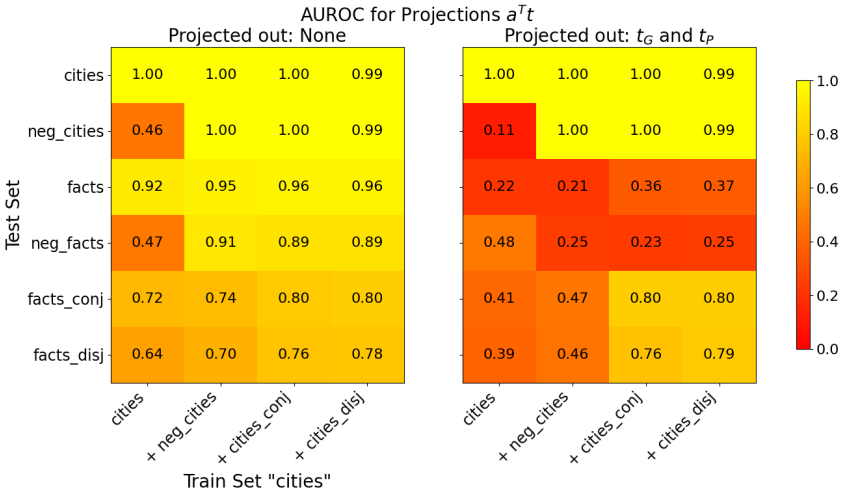
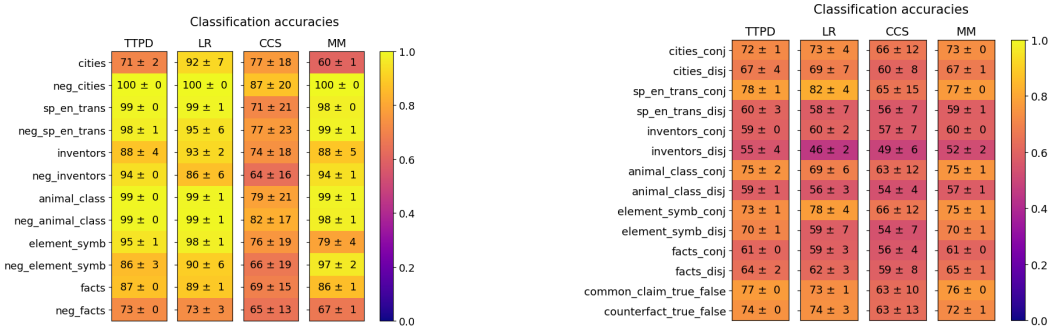


Figure 37: Llama3-8B-base: Generalisation accuracies of truth directions $\mathbf{t}$ before (left) and after (right) projecting out $\mathbf{t}_G$ and $\mathbf{t}_P$ from the training activations. The x-axis shows the train set and the y-axis the test set. All truth directions are trained on 80% of the data. If test and train set are the same, we evaluate on the held-out 20%, otherwise on the full test set. The displayed AUROC values are averaged over 10 training runs, each with a different train/test split.



(a) (b)

Figure 38: Llama3-8B-base: Generalization of TTPD, LR, CCS and MM. Mean and standard deviation are computed from 20 training runs, each on a different random sample of the training data.